

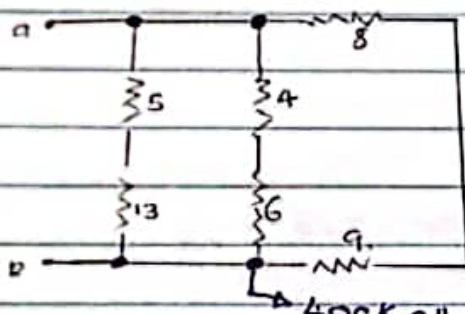
SSC Notes - 07/08/23

Req



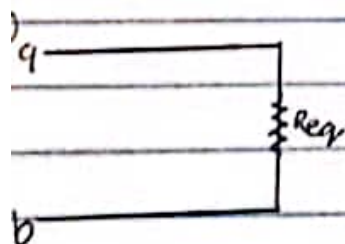
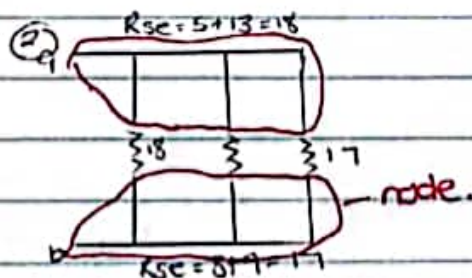
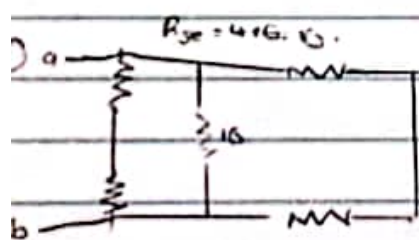
$$R_{eq} = 8 + 6 + 7 = R_{eq}$$

Resistors in series - ONE CURRENT FLOW



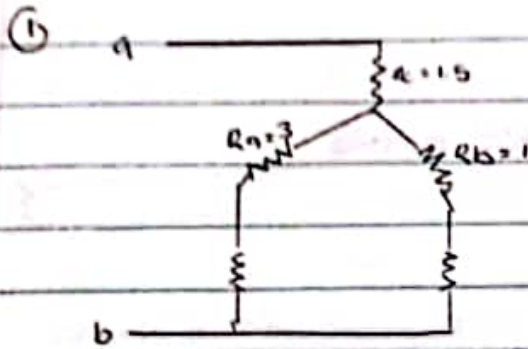
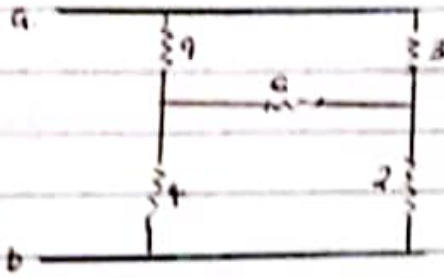
Req between a and b.

Lock all points



$$R_{eq} = \left(\frac{1}{18} + \frac{1}{10} + \frac{1}{7} \right)^{-1}$$

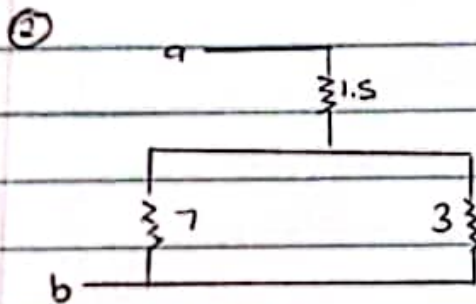
PARALLEL - TWO OR MORE BRANCHES MUST SHARE TWO NODES.



$$R_a = \frac{9 \times 6}{9 + 6 + 3} = 3$$

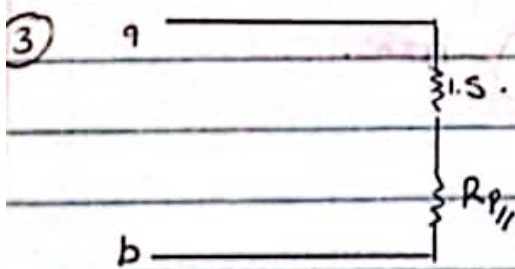
$$R_D = \frac{3 \times 6}{9 + 6 + 3} = 1$$

$$R_c = \frac{9 \times 3}{9 + 6 + 3} = 1.5$$



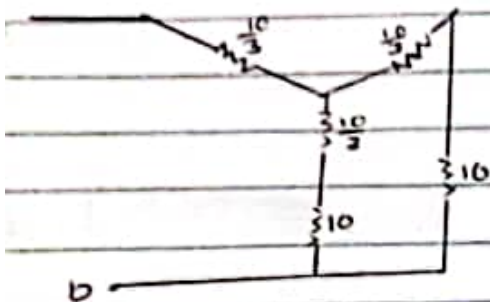
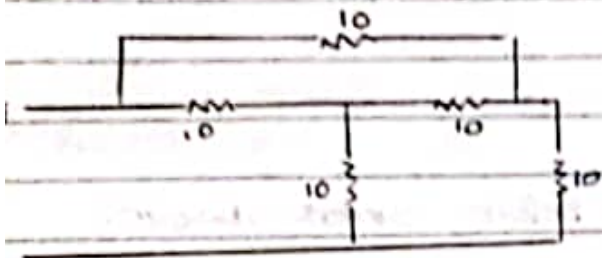
$$R_{se} = 3 + 4 = 7$$

$$R_{se} = 1 + 2 = 3$$

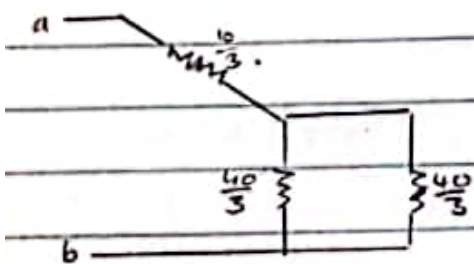


$$R_{p||} = \left(\frac{1}{7} + \frac{1}{3} \right)^{-1}$$

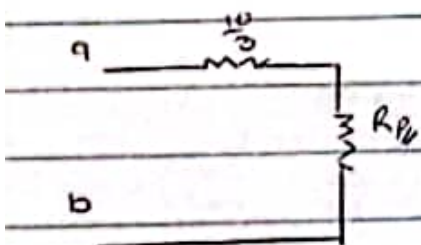
$$R_{eq} = 1.5 + R_{p||}$$



$\frac{10}{3}$ } if all 3 resistors have the same value
 $\frac{10}{3}$



$$R_{sc} = \frac{10}{3} + 10 = \frac{40}{3}$$



$$R_{p||} = \left(\frac{40}{3} + \frac{40}{3} \right)^{-1}$$

$$R_{eq} = \frac{10}{3} + R_{p||}$$

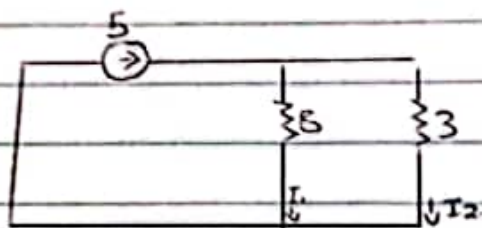
CURRENT DIVISION

KCL - Current division

KVL - Voltage division

Always try to use KCL before current division.

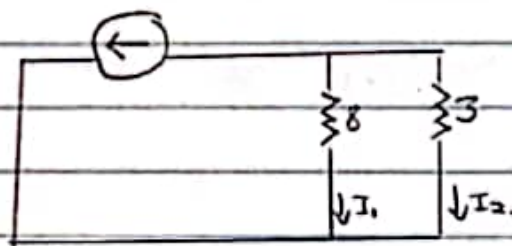
CURRENT DIVISION - ONLY IF TWO RESISTORS ARE SHARING TWO NODES.



$$I_1 = \left(\frac{\text{opposite res}}{\text{sum of res in ||}} \right) \times \text{current}$$

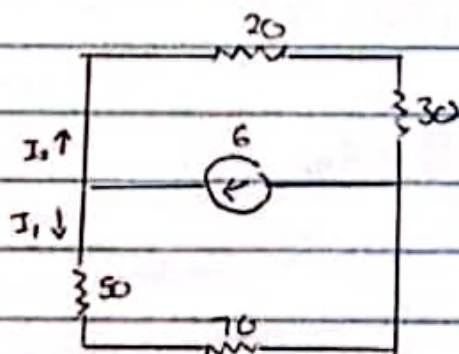
$$= \left(\frac{3}{8+3} \right) \times 5$$

$$I_2 = \left(\frac{8}{8+3} \right) \times 5$$



$$I_1 = \left(\frac{3}{8+3} \right) \times -5$$

$$I_2 = \left(\frac{8}{8+3} \right) \times -5$$



$$I_1 = \left(\frac{(20+30)}{(20+30+50+10)} \right) \times 6$$

$$I_2 = \left(\frac{(50+10)}{(20+30+50+10)} \right) \times 6$$